



FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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CORE SKILLS

Programming: C++, C#, Python

Graphics & Shaders: HLSL, GLSL, Compute Shaders, VFX Graph

Engines & Tools: Unity (Expert), Unreal Engine 5, Custom OpenGL Frameworks, Blender

Workflows: Git, Jira, Project Management, Agile Product Strategy

EDUCATION

M.Sc. Cinema and Media Engineering

Politecnico di Torino | 2024 - 2026
Expected mark: 110/110 Cum Laude (Dec 2026).

B.Sc. Cinema and Media Engineering

Politecnico di Torino | 2021 - 2024
Final mark: 100/110.

PERSONAL PROJECTS

Smart Cable Gen (Blender Add-on)

Python / Pipeline Tooling

Developed a procedural multi-object cable authoring add-on in Python with scale-independent catenary physics and custom UI for game-ready export.

Graphics Lab (Custom C++ Engine)

Engine Programmer

Engineered a custom OpenGL 3D rendering framework from scratch with Deferred Shading, PBR, and SPOM.

UE5 Tech Art & Python Tools Procedural Workflows & Rigging

Built procedural environments (PCG) and custom FK/IK character rigs. Developed Python scripts for pipeline automation and integrated MetaHumans.

Awaken Asylum & Nihil

Lead Technical Artist

Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype. Managed end-to-end technical production.

PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

WORK EXPERIENCE

XR Master Thesis - R&D Graphics Programmer 03/2026 - Present

LEMONS IN THE ROOM & POLITECNICO DI TORINO

Torino / Firenze, Italy

- Engineered **PulseEngine**, a real-time bio-adaptive telemetry and volumetric CSG framework in Unity 6 for standalone Medical XR (Meta Quest 3).
- Architected a **Virtual DAW Mixer Console** with live oscilloscopes and auto-sniffing to route multi-sensor biometric streams (BLE, OSC, WebSockets).
- Eliminated triplanar symmetry seams on SDFs via **Morten Mikkelsen Surface Gradients** and authored a GPU Voxelizer (<0.8ms) powering VFX Graph particle swarms.
- Implemented **Physics-Only Mode** to decouple GPU fill-rate, locking **90 FPS on standalone Quest 3** (61.3% GPU time reduction, 0B GC Alloc).

Technical Artist & Project Manager

01/2025 - 07/2026

LEVEL UP LAB (POLITO STUDENT TEAM)

Torino, Italy

- Programmed core gameplay mechanics in C# (Unity) and designed the technical architecture for the "Line Operator" project.
- Acted as Project Manager and Technical Producer, coordinating multidisciplinary teams of designers and artists to oversee artistic vision and technical viability.
- Developed custom HLSL Shaders and post-processing pipelines to establish the game's unique visual identity.

Teaching Assistant (GUI)

10/2024 - 01/2025

POLITECNICO DI TORINO

Torino, Italy

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

Video Artist

09/2023 - 06/2024

RKH SRL

Torino, Italy

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.